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Interesting Problems from Class

These are problems people frequently asked me about during class or sent emails for. I thought I'd type up solutions, so everyone could read and understand these better with examples.

Question 1. Stated in Figure 1.

When a basketball player makes a trip to the free throw line, he takes two consecutive shots. It is often wondered whether these two shots are independent or dependent: does the probability of making the second free throw *depend* on whether a player makes the first free throw?

After analyzing data for basketball star Albert Jones, statisticians determined that his first and second free throws are entirely *independent events*. The frequency table below shows the data that analysts used to determine this independence.

Fill in the missing values from the frequency table.

	Makes first shot	Misses first shot	Row total
Makes second shot			180
Misses second shot			70
Column total	200	50	250

FIGURE 1—Problem from Jay on August 26th

The assumption that does the heavy-lifting here is of *independence*. This means if we can somehow tell the chances or odds of the player making a shot and missing a shot, we can uniformly apply them trial after trial, i.e., shot after shot. Treating each new shot, as the first one. Since, the previous shot has no bearing over the next one.

So what are the chances of making a shot or missing a shot? We may calculate that from the first shot successes.

There were,

$$\frac{200}{250}\ \text{Makes} \qquad \frac{50}{250}\ \text{Misses}$$

This tells us that the player is roughly making 4/5 shots and missing 1/5 shots.

Per such estimate, if they made 180 more shots, they will,

$$\text{Make}\quad \frac{4}{5}\cdot 180 = 4\times 36 = 144, \qquad \text{Miss}\quad 180-144=36.$$

From here the easiest way to figure out the second row is to count up to the column totals, e. g., $200-144=56$ and so on.

Question 2. Stated in Figure 2.

The key observation here is to correlate a quantity to it's percentage in order to figure out the total.

For instance, we know there are 48 squares, and they make up 65.75% of the total.

$$\frac{48}{\text{total}}\cdot 100 = 65.75$$

Or similarly, we know there are 33 off-white tiles making up 45.32% of the total.

$$\frac{33}{\text{total}}\cdot 100 = 45.32$$

Tyler is bored in history class, so he is staring at the ceiling tiles. He notices that some tiles are perfect squares, while others are long rectangles. He also notices that some of the tiles are bright white, while others are off-white. He makes detailed notes on how often each of these tiles occurs.

Tragically, Tyler crumpled up his notes when the teacher looked in his direction, and now he can't read all the numbers. The absolute and relative frequency tables below show the numbers that Tyler is able to read. Can you help him figure out the rest?

		Square	Rectangle	Row total
Bright white	Row %	62.50	37.50	100
	Column %	52.08	60.00	------
	Total %	34.25	20.55	54.79
Off-white	Row %	69.70	x	100
	Column %	47.92	40.00	-----
	Total %	31.51	13.70	45.21
Column total	Row %	-----	-----	100
	Column %	100	100	100
	Total %	65.75	34.25	100

Fill in the missing values from each table.

	Square	Rectangle	Row total
Bright white	25		
Off-white	23	10	33
Column total	48		

$x =$

FIGURE 2—Problem from Eli on August 26th

In each case, the total (Row total or Column total) is rounded to 73. Once you know that, the rest of the columns/rows could be counted up to 73, e. g., $73 - 33 = 40$ and therefore the row total of bright white is 40. The value of x is just counting 69.7 upto the column total of a 100. Hence, solve, $69.7 + x = 100$.

Question 3. Stated in Figure 3.

The table below describes phone calls made on the planet K-2L last year. The columns show calls dialed by men versus women, and the rows show the distribution of call lengths. (Note that this is not real data.)

There were 13,950,000,000 phone calls of 0 to 5 minutes.

Length of call		Men	Women	Row total
0 to 5 minutes	Row %	66.4	33.6	100
	Column %	75.0	75.0	------
	Total %	49.8	25.2	75.0
5 to 30 minutes	Row %	66.4	33.6	100
	Column %	20.0	20.0	-----
	Total %	13.28	6.72	20.0
30 or more minutes	Row %	66.4	33.6	100
	Column %	5.0	5.0	-----
	Total %	3.32	1.68	5.0
Column total	Row %	-----	-----	100
	Column %	100	100	100
	Total %	66.4	33.6	100

How many phone calls were made by women?

Consider 3 particular phone calls from last year:

- The first call lasted 2 minutes.
- The second call lasted 17 minutes.
- The third call lasted 4 hours and 11 minutes.

Which of the following are true?
Choose all answers that apply:

- A The second call is more likely to be made by a man than by a woman.
- B The second call is more likely to be made by a man than either the first or third call.
- C The third call is more likely to be made by a man than either the first or second call.
- D The first call is more likely to be made by a man than either the second or third call.
- E All three calls are equally likely to be made by a man.

FIGURE 3—Problem from Bryan on August 26th

We know that there were 13,950,000,000 calls made that lasted 0–5 minutes and this is 75% of the data.

$$\frac{13,950,000,000}{\text{total}} \cdot 100 = 75$$

Therefore, the total number of calls were 18,600,000,000. We know that 33.6% of this quantity is 6,249,600,000 and as the table tells us, this is the percentage of total calls made by woman.

Note that the percentage breakdown for all rows is 66.4+33.6 = 100. Therefore, the option E applies. If E applies then none of the B,C and D apply. A applies since 66.4 > 33.6 across the durations.

The trap in B/C/D is reading the "Column %" or "Total %" sub-rows instead. Those tell you the second call is rarer than the first, but "rarer" isn't the same as "less likely to be male." The question is about gender given length, which is what "Row %" gives you.

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